

VERESHCHAGINO KRASNOYARSK, Russian Federation

WMO: 235780

Lat: 64.236N

Lon: 87.573E

Elev: 30

StdP: 100.96

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.6	(c) -42.9	(d) -48.1	(e) 0.0	(f) -45.6	(g) -45.6	(h) 0.0	(i) -43.0	(j) 7.5	(k) -16.6	(l) 6.4	(m) -15.2	(n) 0.8	(o) 180	(p) 0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.3	18.2	26.4	17.4	24.2	17.2	20.0	24.9	18.8	23.7	17.8	22.4	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.1	22.5	17.0	12.2	21.0	16.0	11.4	20.4	57.7	25.2	53.9	23.6	50.5	22.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.7	5.2	-47.6	31.1	2.6	2.1	-49.4	32.5	-51.0	33.7	-52.4	34.9	-54.3	36.4		
			-48.0	21.4	2.8	1.7	-50.1	22.7	-51.7	23.7	-53.3	24.6	-55.3	25.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.9	-25.8	-20.6	-10.9	-3.7	2.6	14.9	17.7	13.0	6.4	-3.3	-18.4	-20.3
	DBStd	16.16	10.14	8.47	7.95	6.32	4.99	5.29	3.77	3.85	4.40	5.82	9.19	9.96
	HDD10.0	5624	1108	857	649	411	234	17	1	16	126	413	851	939
	HDD18.3	8195	1367	1091	908	661	487	127	58	168	359	671	1101	1198
	CDD10.0	536	0	0	0	0	7	164	238	109	18	1	0	0
	CDD18.3	64	0	0	0	0	1	23	37	3	0	0	0	0
	CDH23.3	601	0	0	0	0	3	243	332	22	1	0	0	0
	CDH26.7	132	0	0	0	0	1	56	75	1	0	0	0	0
Wind	WSAvg	2.3	2.2	2.3	2.5	2.6	2.5	2.2	2.1	2.0	2.1	2.4	2.0	2.4
Precipitation	PrecAvg	790	53	43	48	50	58	60	68	84	73	105	78	69
	PrecMax	997	112	80	84	76	99	139	134	168	132	184	136	112
	PrecMin	521	9	17	7	20	25	21	29	25	25	30	33	21
	PrecStd	134	24	15	20	15	19	24	26	31	27	34	25	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.8	-0.9	5.6	11.4	20.9	30.3	30.9	25.7	22.1	11.8	0.6	-1.1
		MCWB	-5.5	-1.6	1.1	5.3	12.7	18.9	19.2	18.3	14.7	8.0	0.1	-1.6
	2%	DB	-6.5	-4.3	2.9	8.7	16.6	28.4	28.9	23.7	18.0	7.1	-1.4	-2.7
		MCWB	-7.2	-5.1	-0.1	3.2	9.8	17.7	18.8	17.2	13.4	5.7	-2.0	-3.2
	5%	DB	-9.1	-7.4	1.3	6.5	12.5	26.5	27.0	21.7	14.9	5.2	-3.0	-4.2
		MCWB	-9.7	-8.0	-0.4	2.3	7.0	16.6	18.0	16.5	11.1	3.9	-3.5	-4.7
	10%	DB	-11.7	-9.7	-0.2	4.5	9.7	24.0	25.1	19.8	12.6	3.1	-5.0	-7.1
		MCWB	-12.2	-10.3	-1.7	1.2	5.7	16.6	17.6	15.7	10.2	1.8	-5.5	-7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.3	-1.5	1.7	5.8	13.8	20.2	21.8	20.2	16.1	8.1	0.2	-1.5
		MCDB	-4.9	-1.0	3.5	8.6	19.5	27.3	27.5	24.0	19.7	10.8	0.5	-1.3
	2%	WB	-7.2	-4.9	0.7	4.1	10.4	18.9	20.7	18.3	13.6	6.1	-1.9	-3.3
		MCDB	-6.5	-4.4	2.3	7.3	15.8	25.2	26.5	21.4	17.4	6.9	-1.5	-2.9
	5%	WB	-9.7	-7.9	-0.3	2.8	8.1	18.0	19.6	17.1	12.0	4.1	-3.6	-4.7
		MCDB	-9.2	-7.4	1.3	5.8	11.7	23.6	24.2	20.7	14.1	5.0	-3.2	-4.1
	10%	WB	-12.2	-10.2	-1.6	1.6	6.2	17.0	18.6	16.1	10.5	2.1	-5.6	-7.5
		MCDB	-11.7	-9.7	-0.2	4.1	8.9	22.8	23.1	19.5	12.3	2.9	-5.1	-7.0
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.5	9.9	9.6	7.6	10.3	9.3	8.3	6.3	5.0	8.1	7.5
		MCDBR	6.0	10.1	8.6	10.5	12.6	14.2	13.5	11.9	10.0	6.3	7.4	6.2
		MCWBR	5.8	10.0	6.6	7.1	7.3	6.1	5.9	6.7	5.9	4.6	7.6	6.0
	5% WB	MCDBR	6.5	10.3	8.2	9.0	11.0	10.7	10.8	10.2	8.2	5.6	7.0	5.9
		MCWBR	6.3	10.3	7.0	6.6	7.1	5.7	5.9	6.6	5.6	4.6	7.1	5.9
Clear-Sky Solar Irradiance	taub	0.234	0.284	0.326	0.349	0.367	0.383	0.429	0.408	0.347	0.299	0.216	0.226	
	taud	1.874	1.989	1.991	2.038	2.210	2.386	2.232	2.261	2.402	2.196	1.904	1.867	
	Ebn at Noon	279	586	726	809	837	834	773	750	733	581	296	87	
	Edh at Noon	30	70	108	130	121	105	119	105	74	56	27	12	
All-Sky Solar Radiation	RadAvg	0.18	0.94	2.51	4.41	5.40	5.63	4.91	3.30	1.90	0.75	0.29	0.07	
	RadStd	0.02	0.07	0.18	0.44	0.51	0.49	0.55	0.42	0.26	0.12	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.98	N/A	N/A	N/A	N/A	N/A	-372	-455	N/A	N/A

Nomenclature: See separate page